

KAPIL RAVI RATHOD

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EDUCATION

The Pennsylvania State University, University Park

GPA: 3.7 | M.Sc. in Computer Science and Engineering

Aug 2024 – May 2026 (Expected)

State College, PA

National Institute of Technology, Warangal

B.Tech. in Computer Science and Engineering

Aug 2017 – May 2021

Warangal, TS

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, Go, R, JavaScript, Bash, CUDA C++

Databases: MySQL, PostgreSQL, Oracle, MongoDB, Redis, Qdrant, FAISS, Milvus

Dev Tools: Spring Boot, Angular.js, Node.js, React.js, Next.js, Django, Docker, Git, Hadoop, Kafka, Spark, gRPC

Testing & Cloud: PyTest, Unittest, Locust, Behave, Selenium, JUnit, GCP [Link], Prefect, OpenShift, Jenkins, Kubernetes

AI/ML Tools: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, LangChain, AutoGen, LangGraph, DeepEval, Google ADK

EXPERIENCE

Recursion

May 2025 – Aug 2025

SWE Intern

Salt Lake City, UT

- Refactored and shipped **4 Django and React/Next.js components** for the Experiment Configuration orchestration engine.
- Streamlined Prefect pipelines, automating data ingestion to GCP cloud storage, **improving operational efficiency by 60%**.
- Supported **3 critical** dashboards (Hex, Looker) through diverse visualizations, enabling data-driven insights from RADME data.
- Contributed to a multi-agent chatbot suite integrating with platforms like ServiceNow and Splunk to automate lab processes, diagnose incidents and design experiments, **reducing manual effort by 50%**.

Deutsche Bank

Jul 2021 – Jul 2024

SDE2 - Associate Software Engineer

Pune, MH

- Architected a self-service data ingestion pipeline to automate the extraction, semantic chunking and vectorization of Confluence and SharePoint documents into Qdrant for **100+ teams**.
- Engineered an LLM-based alerting & monitoring service for global regulatory tracking, **saving \$1M/year and 700+ hours**.
- Implemented a Plotly dashboard for **100+ users** to monitor and manage DaS platform computational resource usage per user.

SDE1 - Graduate Analyst

- Automated UC4 pre-deployment checks with Python/UNIX to parse config files, **reducing manual effort by 90%**.
- Enhanced a bank-wide translation service for intranet webpages, **boosting translation quality (BLEU scores) by 12%**.
- Improved a reporting app's infra, security and UI; created dashboards that **saved 30+ hours/week** on manual updates.
- Built a BERT-based automated resume matching tool, **saving 80% of manual effort across 100+ hiring managers**.

Semut.io

Apr 2020 – Jul 2020

Cloud Native Engineering Intern

Remote

- Contributed to a Go-based application manager by implementing **3 core features** for Elasticsearch-as-a-service provisioning.

HERE Technologies

May 2019 – Jul 2019

SWE Intern

Mumbai, MH

- Designed a UI-based RESTful bulk-upload POI spell-checking service **saving 100+ hours** for the map creation team.
- Ideated a POC for stitching Here vehicle Lidar images, enhancing vehicle perception by processing **1,000+ frames**.

PROJECTS

ClaimShield - NexHacks Carnegie Mellon University Hackathon 2026

Jan 2026 - Jan 2026

- Built an AI-powered healthcare advocacy platform that uses LLMs to audit clinical notes against Medicare standards, automatically detecting billing anomalies and generating formal appeal letters.

Intelligent Autonomous Knowledge Navigator

Oct 2025 – Dec 2025

- Integrated voice control, RAG and computer vision into a multi-agent Python application capable of executing autonomous research and desktop automation tasks.

Northropp Grumman Object Detection Hackathon 2024

Nov 2024 – Nov 2024

- Designed an award-winning real-time object detection system with YOLOv7, Grounding DINO, and an interactive UI.

Composed Image Retrieval with Contrastive & CLIP-based Features

Oct 2024 – Dec 2024

- Improved image retrieval performance on the FashionIQ dataset by redesigning the CLIP-based Combiner with multi-head attention and scalar weighting, utilizing a two-stage training approach.

InfraAlert - Deutsche Bank AI/ML Hackathon 2023

Jun 2023 – Jul 2023

- Built a system for resource (CPU/RAM/Disk/NetPacket) prediction and real-time anomaly detection, utilizing LSTM/CNN-LSTM for forecasting, CTGAN for data synthesis, and XGBoost for classifying streaming network data.

PUBLICATIONS

K.R. Rathod, S. Bansal, Y.H.C. Kandpal and M. Bisi *Learning Automata-based Feature Selection for Duplex Output Software Effort Estimation Model*, 12th ICCNT, Kharagpur, India, 2021. [Link]